STATISTICAL INFERENCE

Statistical Investigation of Nepal: Development Indicators, 2010–2024

ASSIGNMENT 1

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DECLARATION

I hereby declare that the project work entitled **"Statistical Investigation of Nepal: Development Indicators, 2010–2024"** submitted to the department is an authentic record of my own work carried out as part of the course requirements. I declare that I used AI assistance to support the development and understanding of the code for this assignment. I used this assistance mindfully and critically, reviewed the code and report, and checked the consistency of the results presented across them. I take full responsibility for the submitted work, including any errors or omissions in the code, analysis, or document.

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Abstract

As part of this study, statistical inference modeling was performed on empirical data from Nepal using World Development Indicators from the World Bank covering 2010–2024 across nine development metrics. Descriptive statistics, time-series analysis, distributional plots, dispersion measures, and correlation analyses document structural trends over more than a decade. Key findings show that GDP per capita, Internet penetration, and essential service access indicators increased overall, whereas GDP growth exhibited significant volatility and contracted in 2020 during the COVID-19 pandemic. Relative variation was highest for Internet adoption, whereas forest area remained constant across observed data. GDP per capita demonstrated strong positive correlations with life expectancy, electricity access, and CO_2 emissions. Two confidence intervals and hypothesis tests illustrate the t-method, though inferential interpretations remain conditional on assumptions that may be constrained by consecutive annual time-series observations. Missing data points are retained per project requirements, variable substitutions are explicitly disclosed, and associations are distinguished from causal effects. The code is available at Github: https://github.com/rbhdsks/Statistical-Inference-Assignment-2026

Keywords: Heterogeneity, Missingness, Covariance, Confounding, Confidence Intervals

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1 Introduction

Development involves changes in economic output, health, education, infrastructure, technology and environmental conditions. These dimensions need not improve together or at the same rate. This investigation examines nine development indicators for Nepal during 2010–2024 using the saved World Bank World Development Indicators dataset (1). Its purpose is to describe the observed changes, compare variability, examine statistical relationships and explain the limits of conclusions supported by the data.

Five questions organise the analysis. How did GDP per capita and GDP growth change? How did access to water, electricity and the Internet evolve? Which selected indicators display the greatest and lowest variation? How is GDP per capita associated with life expectancy, electricity access and emissions per capita? Which missing observations and unusual changes require investigation?

The workflow moves from measurement and data quality to descriptive statistics, graphical analysis, heterogeneity, illustrative inference and correlation. This sequence prevents statistical results from being interpreted without considering their underlying measurements. The observations represent consecutive years for one country, rather than independent draws from a population of countries or individuals. Consequently, temporal trends and possible dependence are central limitations throughout the investigation.

2 Data and Measurement

The working dataset contains one row for each year from 2010 through 2024, three identification fields and nine indicator columns. Of 135 potential indicator-year values, 133 are observed. Secondary enrollment is missing in 2018 and forest area in 2024. Both have 14 observations; each remaining indicator has 15. Missing values remain missing, without interpolation, extrapolation or replacement by zero.

Individual summaries use available observations. Correlations use paired observations, retaining a year only when both measurements exist. All three required correlation pairs contain 15 years, so no observations are excluded from those calculations. Nevertheless, the two incomplete indicators describe slightly different observed periods. Their summaries should not be interpreted as though the missing values were known.

All nine indicators are quantitative and treated as continuous measurements, although their recorded precision differs. Year is a discrete time index, while country name and

country code identify the observational unit. GDP per capita is output per person in constant 2015 dollars, not household income or its distribution. GDP growth is annual percentage change in real output. Life expectancy expresses expected years of life at birth under the mortality conditions represented by the indicator.

Water, electricity and Internet indicators measure percentages of the population. Coverage does not establish reliability, service quality or connection speed. Gross secondary enrollment compares students of all ages with the official secondary-school-age population and may legitimately exceed 100%. Forest area is a percentage of land area; it does not directly measure forest condition or biodiversity.

The assigned CO_2 series could not be downloaded successfully in the project workflow. An explicitly labelled alternative, `EN.GHG.CO2.PC.CE.AR5`, measures CO_2 emissions excluding land use, land-use change and forestry per capita. Its recorded unit is tonnes CO_2 equivalent per capita. Instructor acceptance of this substitution remains unresolved. Raw responses and source metadata are preserved, and structural and plausibility checks examine duplicates, year coverage, country identity, numeric values and appropriate bounds. Passing these checks does not eliminate measurement uncertainty.

The dataset also separates the unit of observation from the population referenced by each measure. For example, an electricity value describes population coverage, but this analysis has annual country-level observations rather than individual household records. Fifteen years therefore does not mean fifteen people were sampled. Similarly, national averages can conceal differences between provinces, urban and rural areas, or income groups. Those differences cannot be recovered from the present dataset. Exact source definitions and download information belong in the accompanying variable dictionary, allowing readers to distinguish documented measurement choices from assumptions made for analysis.

3 Descriptive Statistics

The consolidated table reports all eleven required statistics for each indicator. Sample variance and standard deviation use denominator $n - 1$. This convention specifies a calculation; it does not establish that the observations form a random sample. Quartiles use linear interpolation, and the interquartile range is the difference between the third and first quartiles. Full precision is retained during calculation, with rounding applied only

to displayed results. (The following calculation was done and is commited in my github account for the reference: https://github.com/rbhdsks/Statistical-Inference-Assignment-2026)

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}, \qquad s = \sqrt{s^2}, \qquad \text{IQR} = Q_3 - Q_1.$$

Table 1: Consolidated descriptive statistics for Nepal, 2010–2024.

Panel A: Coverage, central tendency and extremes

Indicator	N	Mean	Median	Minimum	Maximum	Range
GDPpc	15	947.519	951.857	721.265	1179.955	458.690
Growth	15	4.258	4.670	-2.370	8.977	11.347
Life	15	68.542	68.444	66.772	70.636	3.864
Water	15	89.788	89.842	85.762	93.640	7.878
School	14	74.246	73.093	55.589	90.388	34.799
Electricity	15	85.300	89.900	67.300	97.900	30.600
Internet	15	26.458	24.300	7.930	46.322	38.392
CO_2 alt.	15	0.396	0.471	0.175	0.590	0.415
Forest	14	41.591	41.591	41.591	41.591	0.000

Panel B: Dispersion and quartiles

Indicator	Variance	SD	Q_1	Q_3	IQR
GDPpc	23203.39	152.327	823.852	1069.953	246.101
Growth	7.894	2.810	3.473	5.821	2.348
Life	1.470	1.213	67.486	69.203	1.717
Water	6.345	2.519	87.833	91.779	3.946
School	132.190	11.497	66.003	84.057	18.054
Electricity	86.936	9.324	79.700	91.000	11.300
Internet	199.764	14.134	14.370	38.977	24.607
CO_2 alt.	0.023	0.152	0.246	0.526	0.280
Forest	0.000	0.000	41.591	41.591	0.000

Source: saved WDI working dataset. GDPpc: constant 2015 US dollars per person; Growth: annual %; Life: years; Water, Electricity and Internet: population %; School: gross enrollment %; CO_2 alt.: t CO_2e per capita; Forest: land-area %. SD, range and IQR use the indicator's units; variance uses squared units. Differences between percentages are percentage points. Forest variance and SD are effectively zero.

GDP per capita has mean US$947.519 and median US$951.857. Their proximity indicates limited mean–median asymmetry but does not prove a symmetric distribution. The minimum, US$721.265, occurs in 2010, and the maximum, US$1,179.955, occurs in 2024. The mean summarises the observed period's level but conceals the upward trajectory and temporary reversals. It should therefore accompany, rather than replace, the chronological graph.

GDP growth averages 4.258%, below its median of 4.670%. Lower-growth observations pull the mean downward, consistent with the distribution's lower tail. The minimum is −2.370% in 2020 and the maximum 8.977% in 2017. The arithmetic mean assigns equal weight to each year; it is not a compound annual growth rate and does not describe growth in every year.

Internet use has mean 26.458% and median 24.300%, with a minimum of 7.930% in 2010 and maximum of 46.322% in 2024. The mean exceeding the median is consistent with some asymmetry, but sustained expansion is essential to interpretation. A single period average combines early low-coverage years with later high-coverage years and should not be presented as current coverage.

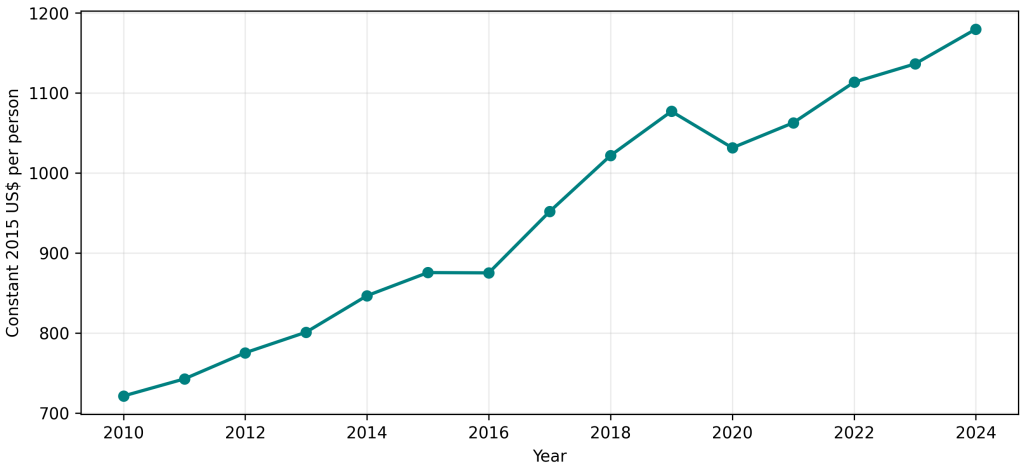
4 Graphical Analysis

4.1 Time-series patterns

The nine time-series plots preserve chronological order, units and missing-data gaps. GDP per capita increases overall, with a small decline in 2016 and a more visible decline in 2020. GDP growth fluctuates considerably, reaching its maximum in 2017 and its only negative observation in 2020. Internet use rises throughout the period, although the final increments are smaller than several earlier increases.

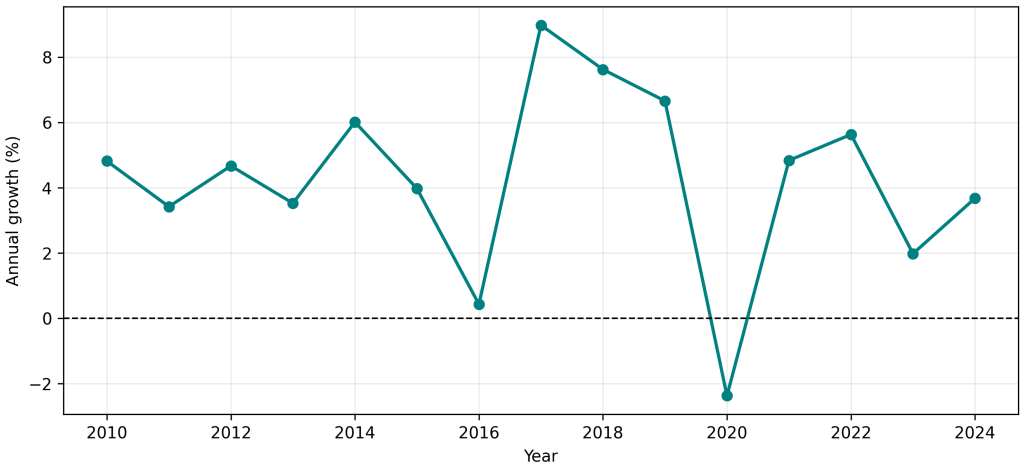
Life expectancy rises overall but falls temporarily in 2015 and during 2020–2021 before recovering. These changes identify timing without establishing causes. Basic drinking-water coverage increases smoothly. Such smoothness does not establish whether each annual value came from a separate direct measurement. Secondary enrollment increases overall, with a missing observation in 2018 and a decline in 2021; the graph does not connect across the missing year.

Electricity coverage expands but includes reversals and a recorded plateau during 2019–2021. The alternative emissions series rises overall, peaks in 2021 and subsequently fluctuates. Forest area is identical across its 14 available observations. This establishes no recorded variation in the saved series, not unchanged ecological conditions. The missing 2024 observation is left unfilled.



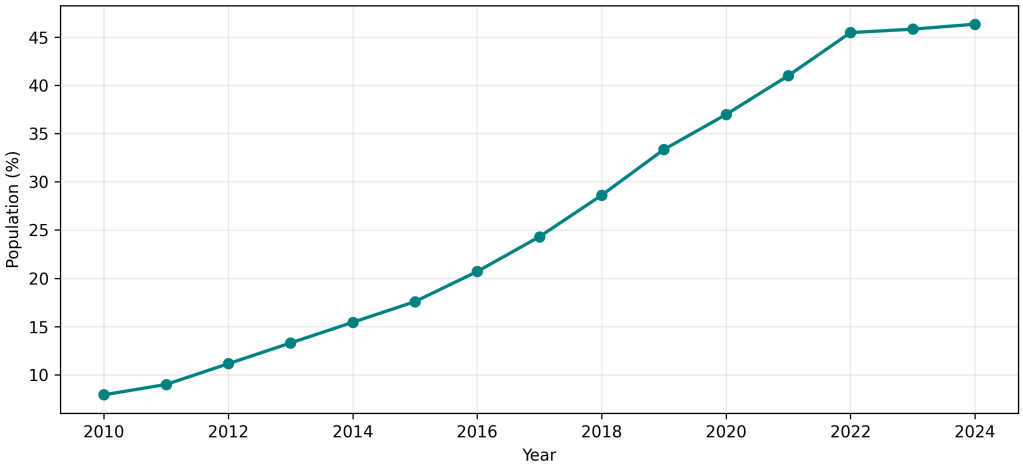
Source: World Bank WDI; saved project dataset.

Figure 1: GDP per capita in constant 2015 US dollars per person.



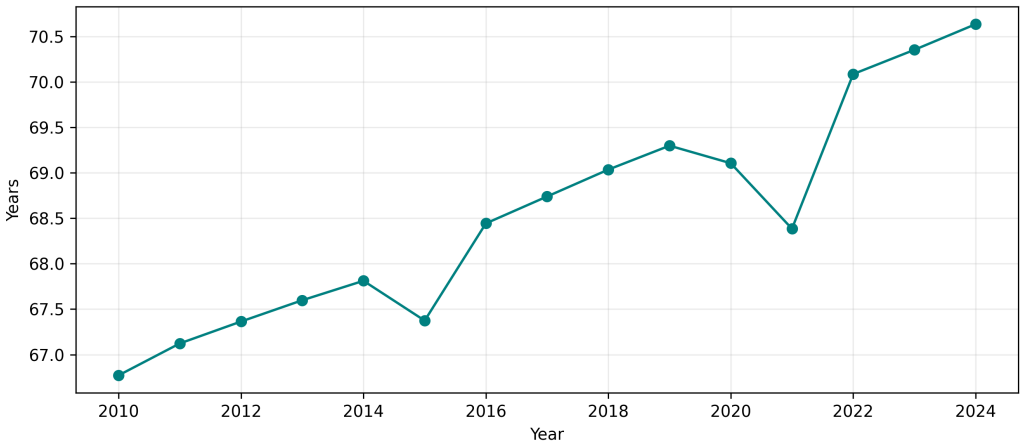
Source: World Bank WDI; saved project dataset.

Figure 2: Annual GDP growth.



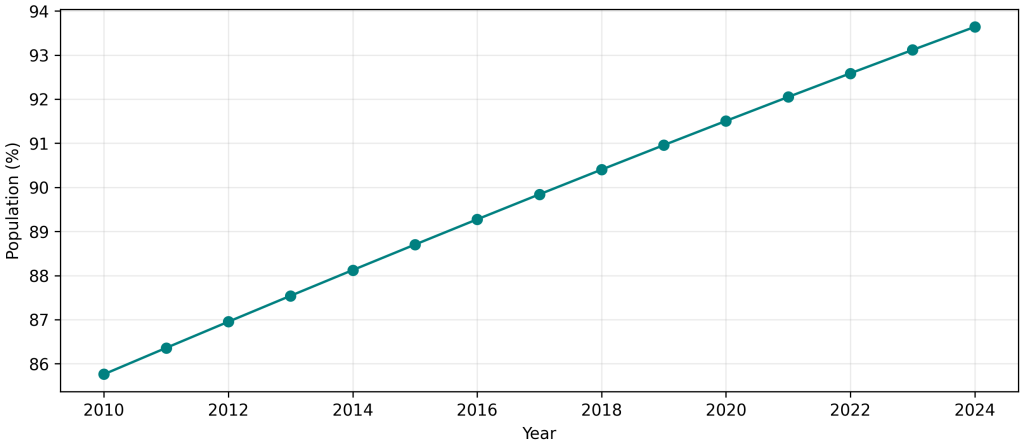
Source: World Bank WDI; saved project dataset.

Figure 3: Individuals using the Internet.



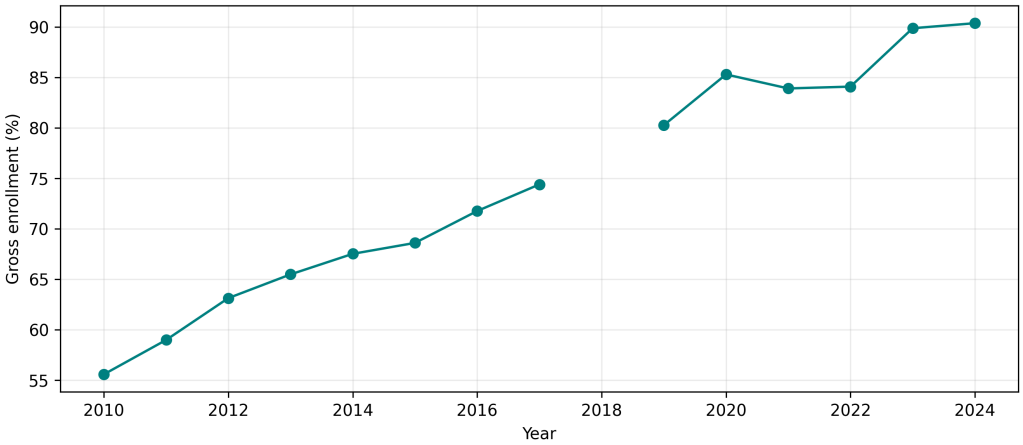
Source: World Bank WDI; saved project dataset.

Figure 4: Life expectancy at birth.



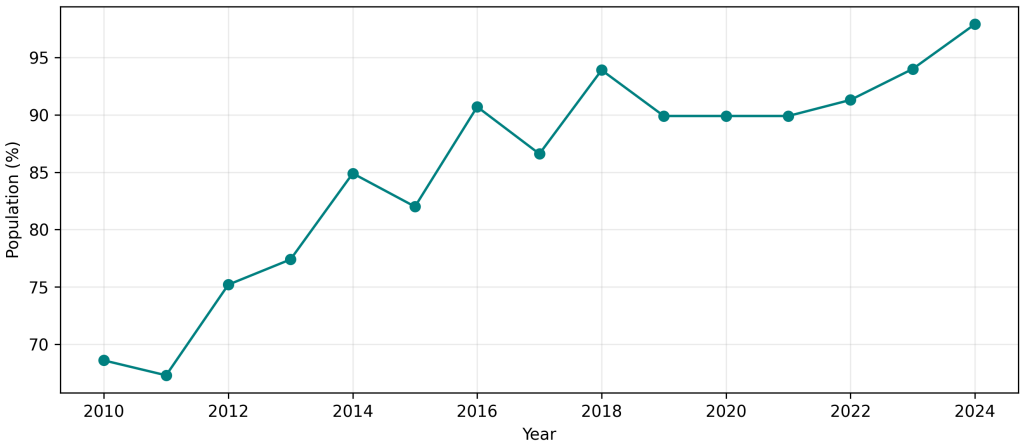
Source: World Bank WDI; saved project dataset.

Figure 5: Population using at least basic drinking-water services.



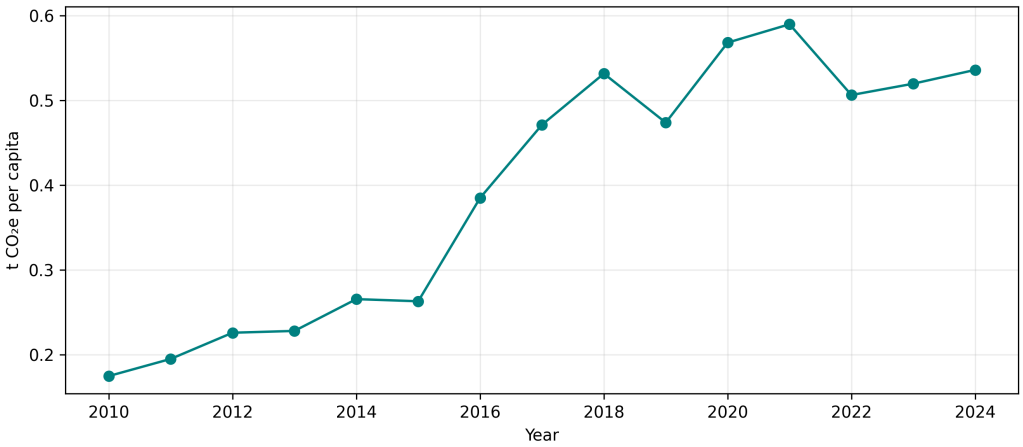
Source: World Bank WDI; saved project dataset. Missing years: 2018.

Figure 6: Gross secondary-school enrollment; 2018 is missing.



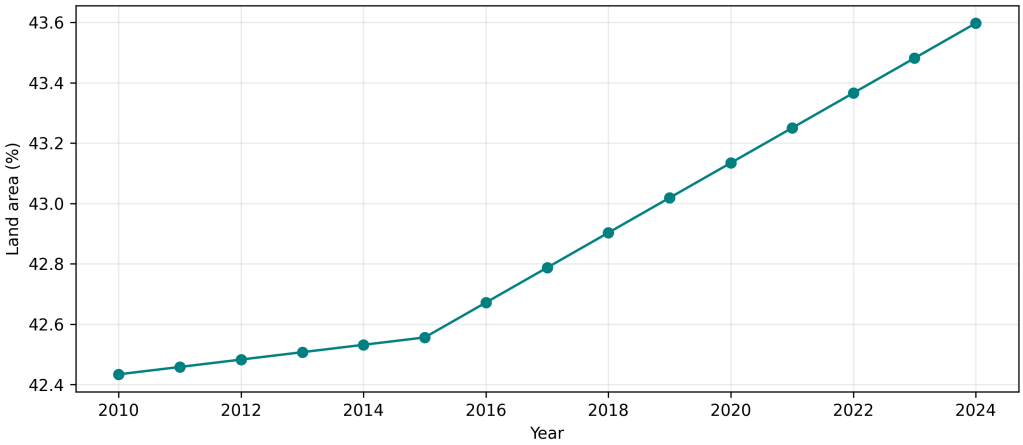
Source: World Bank WDI; saved project dataset.

Figure 7: Population with access to electricity.



Source: World Bank WDI; saved project dataset.
Alternative series: EN.GHG.CO2.PC.CE.AR5.

Figure 8: Alternative CO_2 emissions series, excluding LULUCF, per capita.



Source: World Bank WDI; saved project dataset.

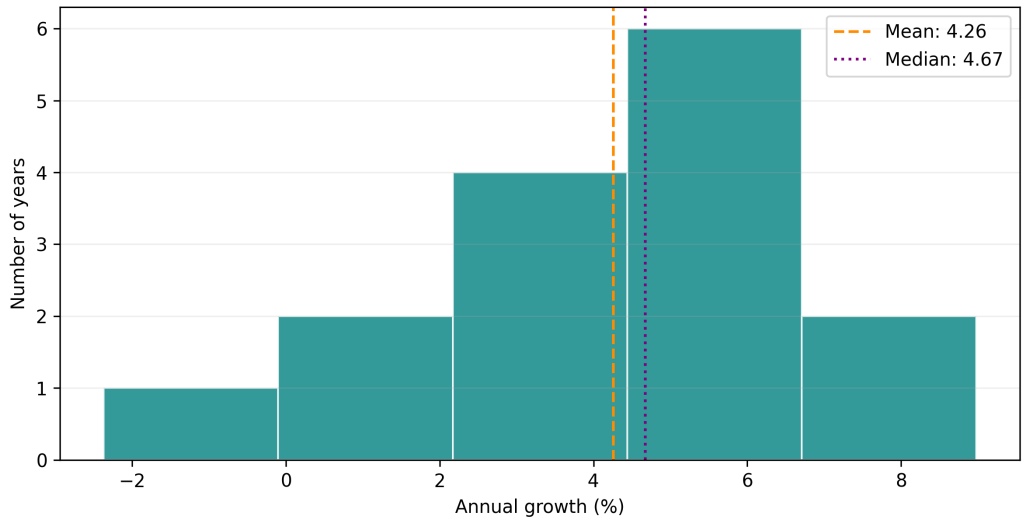
Figure 9: Forest area as a share of land area; 2024 is missing.

4.2 Distributions and unusual observations

GDP growth and Internet use were selected for histograms to contrast a fluctuating indicator with one displaying sustained expansion. Five equal-width bins show predominantly positive growth values with a lower tail, and Internet-use observations spread across different coverage levels. With only 15 observations, apparent shape is sensitive to bin selection; neither histogram establishes normality.

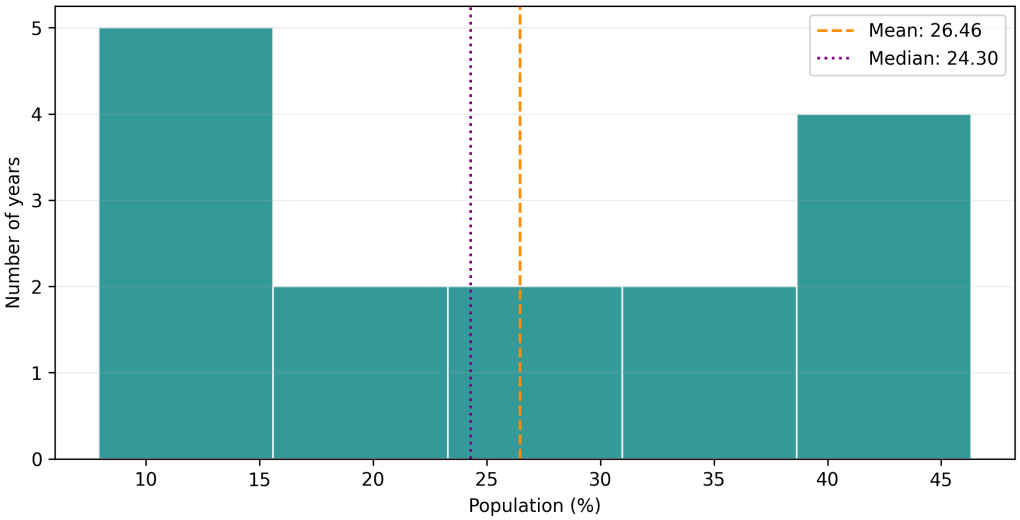
The growth boxplot has median 4.670%, quartiles 3.473% and 5.821%, and IQR 2.348 percentage points. The 2020 observation lies below the lower 1.5-times-IQR fence. Internet use has median 24.300%, quartiles 14.370% and 38.977%, and IQR 24.607 percentage points, with no flagged observations. A flag prompts investigation rather than automatic deletion. Both plot types discard time ordering, so their spread must be interpreted alongside the time-series graphs.

The potential growth outlier receives a separate investigation using surrounding years and an authoritative contextual source. Growth changes from 6.6571% in 2019 to −2.3696% in 2020, then returns to 4.8382% in 2021. A World Bank release describes Nepal's FY2020 contraction in the context of the pandemic (2). This supports treating the observation as consistent with a genuine contraction, while not independently verifying the exact saved value. The publication uses fiscal-year language and an earlier data vintage. The observation is retained unchanged, and the detailed comparison appears in the appendix. This illustrates why graphical detection must be followed by contextual checking rather than mechanical removal.



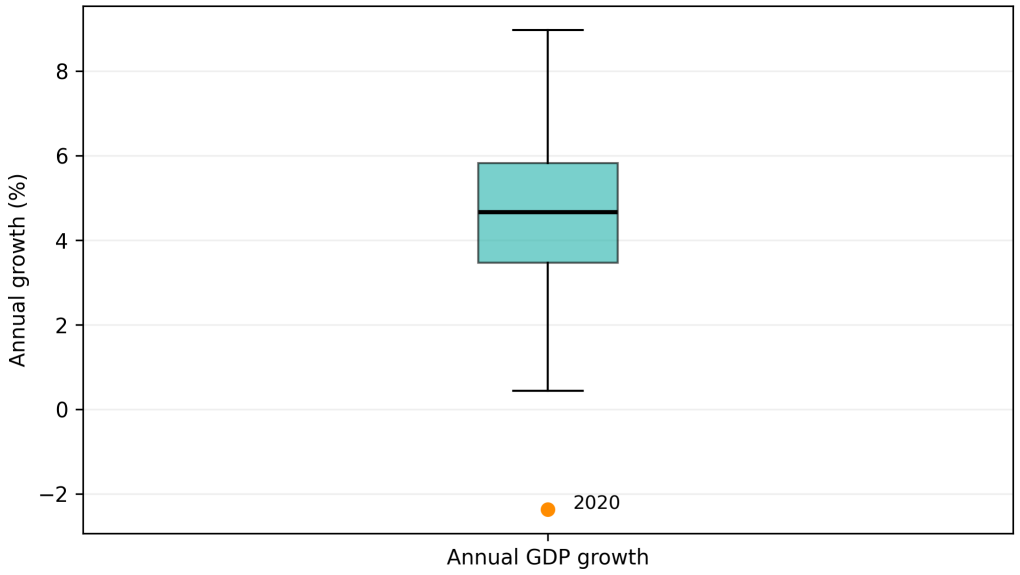
Source: World Bank WDI; 2010–2024; N = 15. Five equal-width bins.

Figure 10: Distribution of annual GDP growth.



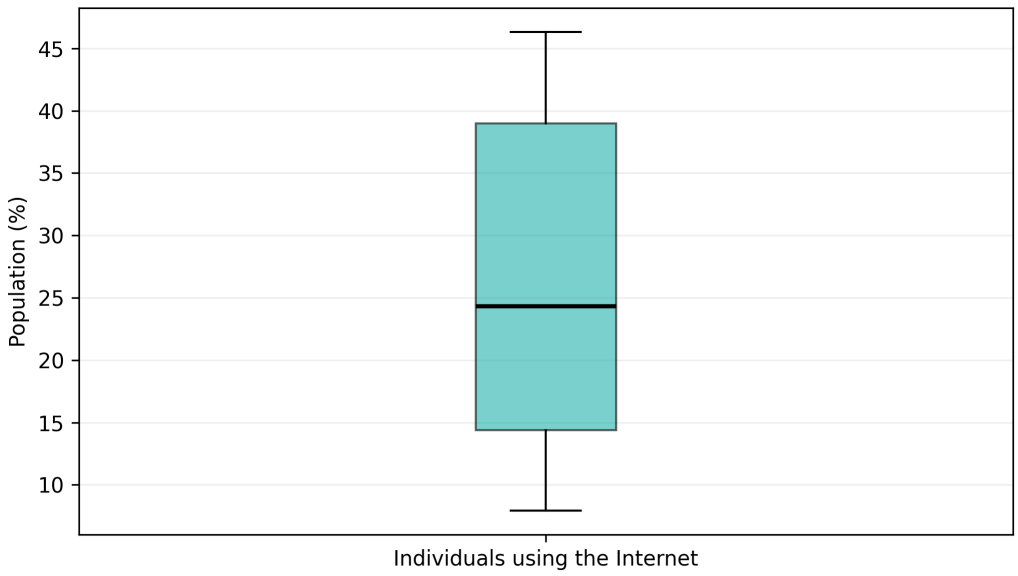
Source: World Bank WDI; 2010–2024; N = 15. Five equal-width bins.

Figure 11: Distribution of annual Internet-use percentages.



Source: World Bank WDI; saved project dataset; N = 15.
Potential outliers identified using the 1.5 × IQR rule.

Figure 12: GDP-growth boxplot with the potential outlier identified by year.



Source: World Bank WDI; saved project dataset; N = 15.
Potential outliers identified using the 1.5 × IQR rule.

Figure 13: Internet-use boxplot.

5 Variance and Heterogeneity

GDP per capita, life expectancy, Internet use and forest area illustrate variation across different units and trajectories. For variance, standard deviation, range and IQR, the numerical ordering is GDP per capita, Internet use, life expectancy and forest area. Under the coefficient of variation, Internet use ranks above GDP per capita, followed by life expectancy and forest area.

Approximate CVs are 53.420%, 16.076%, 1.769% and zero, respectively. CV expresses standard deviation relative to the mean. These selected indicators have positive means and meaningful zero points, supporting a descriptive comparison of relative spread. However, CV does not remove temporal trends, percentage bounds or differences in what the variables measure.

$$\mathrm{CV} = 100\frac{s}{\bar{x}}.$$

Variance and standard deviation necessarily produce identical rankings because square root is a monotonic transformation. Agreement with range and IQR is specific to these observations. Range depends entirely on the extremes, whereas IQR describes the middle half and is less sensitive to isolated extreme values. Neither measure captures the chronological sequence of changes.

Table 2: Heterogeneity across four selected indicators.

Indicator	Variance	SD	Range	IQR	CV (%)
GDP per capita	23203.39	152.327	458.690	246.101	16.076
Life expectancy	1.470	1.213	3.864	1.717	1.769
Internet use	199.764	14.134	38.392	24.607	53.420
Forest area	0.000	0.000	0.000	0.000	0.000

Source: saved WDI working dataset. Units follow Table 1; CV is dimensionless. Tiny floating-point residuals for forest SD are displayed as zero.

Raw variance cannot establish that GDP per capita is substantively more unstable than life expectancy. Expressing GDP in thousands of dollars would divide its variance by one million without changing the underlying observations. Standard deviation in dollars is useful for interpreting GDP spread, with CV providing a relative complement. For Internet use, IQR describes typical coverage spread and CV highlights relative expansion. Life-expectancy SD is interpretable in years. Forest area's zero range clearly communicates no recorded change; tiny computed SD residuals reflect floating-point arithmetic. These statistics describe variation between years, not inequality among Nepal's residents.

6 Confidence Intervals

Two illustrative 95% t-intervals concern hypothetical process means: mean annual GDP growth and mean annual life expectancy. The estimators are the observed averages. The actual averages of the recorded years are directly calculable; interval estimation instead concerns broader parameters under an assumed data-generating process. This distinction prevents ordinary descriptive averages from being mistaken for uncertain estimates of the same fully observed period.

With unknown population standard deviation, the estimated standard error under independent, identically distributed observations is sample SD divided by the square root of sample size. Both indicators have 15 observations, giving 14 degrees of freedom and a two-sided critical value of 2.1448. Multiplying this value by the estimated standard error gives the margin of error.

$$\widehat{\mathrm{SE}}_{\mathrm{IID}}(\bar{x}) = \frac{s}{\sqrt{n}}, \qquad \mathrm{CI}_{95\%} = \bar{x} \pm t_{0.975,n-1}\frac{s}{\sqrt{n}}.$$

The growth standard error is 0.7254 percentage points, producing an interval from 2.7022% to 5.8140%. The life-expectancy standard error is 0.3131 years, producing an

Table 3: Illustrative 95% t-interval calculations.

Indicator	Mean	SD	SE_{IID}	Lower	Upper
GDP growth	4.2581	2.8096	0.7254	2.7022	5.8140
Life expectancy	68.5420	1.2126	0.3131	67.8705	69.2135

Source: saved dataset and t-method calculations. Growth is measured in annual %; life expectancy in years. Margins of error are 1.5559 and 0.6715 respectively. Nominal coverage is conditional on the model assumptions.

interval from 67.8705 to 69.2135 years. Under the specified model, repeated use of this procedure would yield intervals containing the fixed process mean in 95% of repetitions. This is neither a probability assigned to the fixed parameter after calculation nor a prediction interval for next year.

The small-sample method requires independent observations from a common, approximately normal distribution. Independence and normality have not been established, and the life-expectancy trend challenges the common-distribution assumption. These numerical intervals demonstrate the t-method without claiming validated coverage for Nepal's annual series. Their suitability and course coverage must be resolved before final submission.

7 Hypothesis Testing

7.1 Test 1: Mean GDP growth against zero

The first question asks whether the hypothetical process mean of annual GDP growth differs from zero. The null hypothesis is zero mean growth; the two-sided alternative is a nonzero mean. Zero provides a substantive no-growth benchmark rather than a threshold selected to obtain significance. The test uses a significance level of 0.05 and retains all 15 observations, including the 2020 contraction.

$$H_0 : \mu_G = 0, \qquad H_1 : \mu_G \neq 0, \qquad t = \frac{\bar{x} - 0}{s/\sqrt{n}}.$$

Dividing the observed mean's difference from zero by its estimated IID standard error gives $t = 5.8698$ with 14 degrees of freedom. The two-sided p-value is approximately 0.00004076. Under the stated assumptions, the null is rejected. The positive estimate and the corresponding confidence interval excluding zero agree with this conditional decision. They do not establish positive growth in every year or guarantee future growth.

7.2 Test 2: Electricity versus water coverage

The second test compares electricity and basic drinking-water coverage. For each year, electricity percentage minus water percentage defines a paired difference measured in percentage points. The null hypothesis is zero process mean difference, against a two-sided nonzero alternative. All 15 years contain both measurements. Pairing preserves the within-year relationship rather than treating the two sets of measurements as unrelated samples.

$$D_i = E_i - W_i, \qquad H_0 : \mu_D = 0, \qquad H_1 : \mu_D \neq 0, \qquad t = \frac{\bar{D}}{s_D/\sqrt{n}}.$$

The mean difference is -4.4884 percentage points, with SD 7.0994 and estimated IID standard error 1.8331. The resulting statistic is $t = -2.4486$ with 14 degrees of freedom and $p = 0.02812$. At the individual 5% level, the illustrative test rejects zero mean difference under its assumptions. Electricity coverage is lower on average across the observed years, although the 2024 difference is positive. This compares coverage, not the quality of either service.

Pairing does not remove dependence across years or the changing gap over time. Both tests consequently remain illustrative. Each uses 0.05 individually; no family-wise error control is claimed. A p-value is not the probability that the null is true, and statistical significance is not a measure of practical importance. Effect estimates, units and assumptions are therefore reported alongside decisions.

The two comparisons also address different substantive questions: one concerns economic growth relative to no growth, while the other concerns the average gap between services. Rejecting both null hypotheses does not make these effects comparable in size. Their different units and benchmarks must remain explicit. The second result also summarises a changing historical gap; it does not imply that the same gap persists at the end of the period. Reporting the latest observed difference alongside the average helps prevent that misunderstanding.

Table 4: Summary of illustrative hypothesis tests.

Test	N	t	df	p	Conditional decision
Growth versus zero	15	5.8698	14	0.00004076	Reject H_0
Electricity minus water	15	-2.4486	14	0.02812	Reject H_0

Each test uses $\alpha = 0.05$ individually. Decisions are conditional on model assumptions. No family-wise error control across both tests is claimed.

8 Covariance and Correlation

Sample covariance and Pearson correlation are calculated for GDP per capita paired with life expectancy, electricity access and alternative emissions per capita. All pairs contain 15 observations. Covariance uses denominator $n - 1$ and expresses joint movement in the product of the variables' units; Pearson correlation standardises covariance by both standard deviations and is dimensionless.

$$s_{XY} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{n-1}, \qquad r = \frac{s_{XY}}{s_X s_Y}.$$

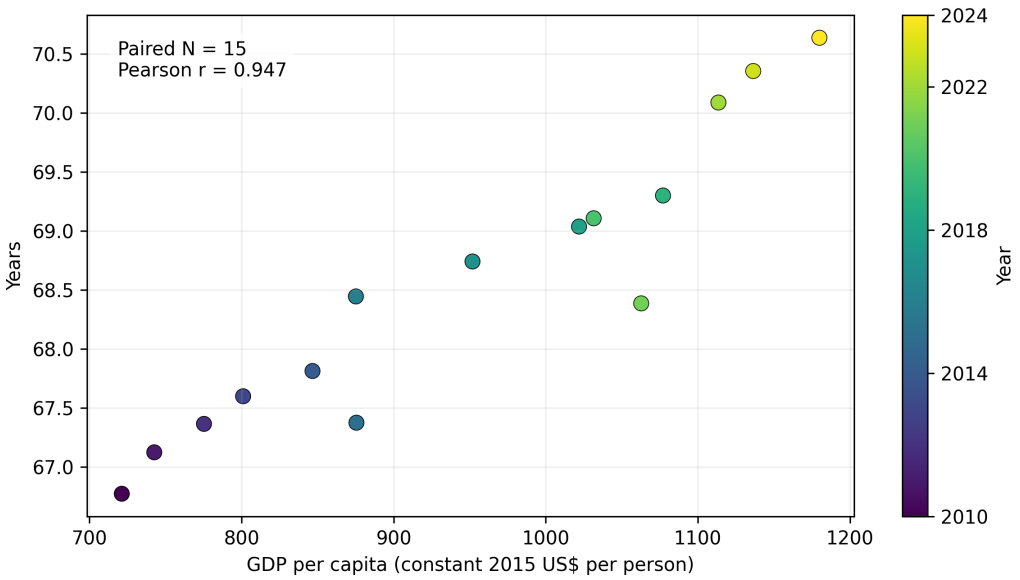
Table 5: GDP per capita paired with three development indicators.

Second indicator	Paired N	Sample covariance	Pearson r
Life expectancy	15	174.8354	0.9465
Electricity access	15	1284.0841	0.9041
Alternative CO_2 per capita	15	21.4271	0.9229

Source: saved WDI working dataset. Covariance units are GDP-per-capita units multiplied by, respectively, years, population-percentage units and t CO_2e per capita. Pearson correlation is dimensionless. No years are excluded from these three pairs.

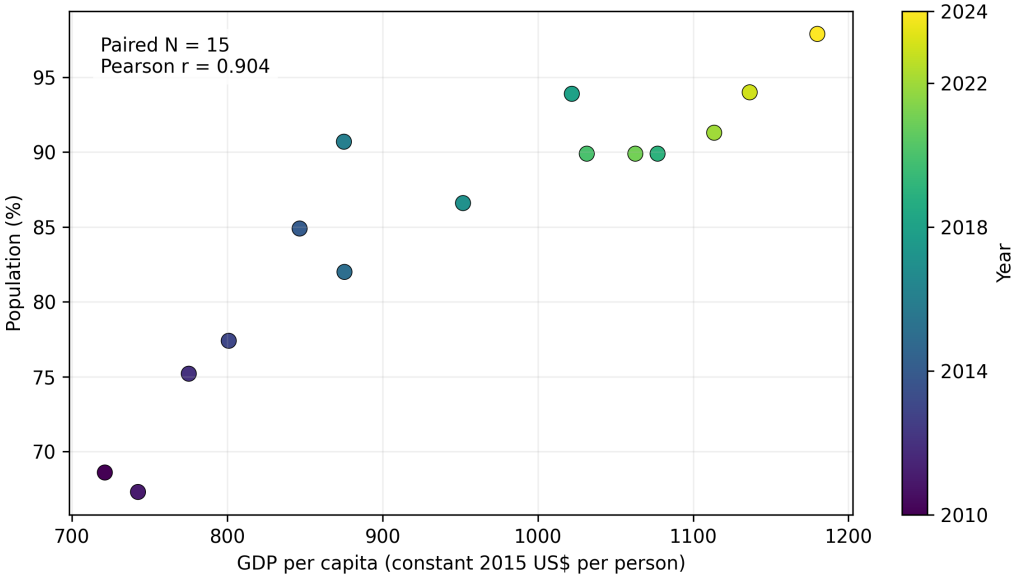
The correlations are 0.9465, 0.9041 and 0.9229, respectively. All indicate strong positive observed linear association. The corresponding covariances are 174.8354, 1284.0841 and 21.4271. Their numerical magnitudes cannot be ranked as association strengths because the units differ. Higher-output years generally coincide with higher values of each comparison indicator.

The scatterplots colour observations by year, making chronological progression visible. Common upward trends can contribute substantially to these correlations; the calculations do not isolate short-run relationships. Their strength cannot establish that increasing GDP causes improvements in health or electricity access, or higher emissions. Correlation significance tests remain omitted pending confirmation of course coverage and a suitable inferential interpretation for dependent annual observations.



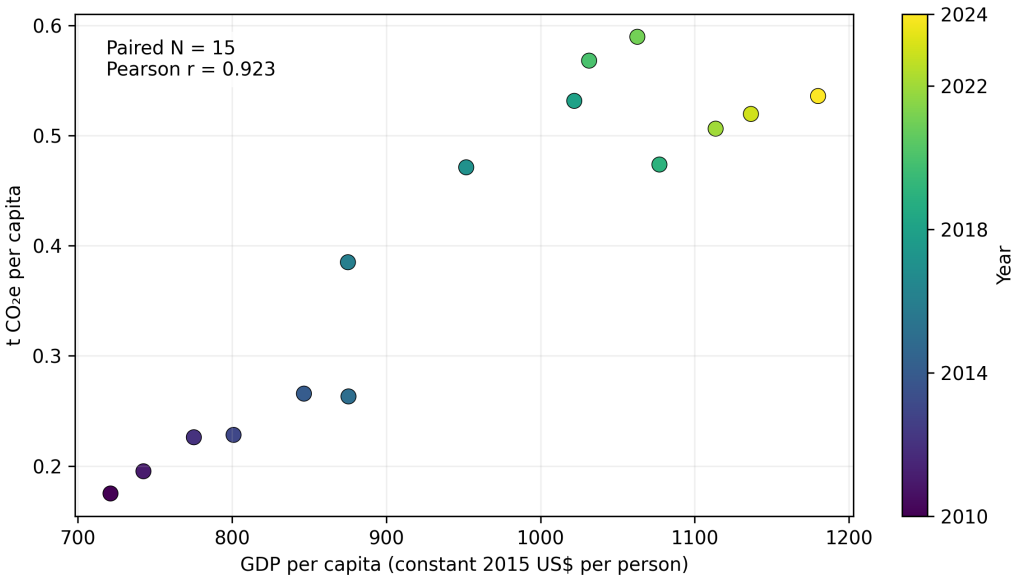
Source: World Bank WDI; saved dataset, 2010–2024.

Figure 14: GDP per capita and life expectancy, coloured by year.



Source: World Bank WDI; saved dataset, 2010–2024.

Figure 15: GDP per capita and electricity access, coloured by year.



Source: World Bank WDI; saved dataset, 2010–2024.
Alternative CO_2 series: EN.GHG.CO2.PC.CE.AR5.

Figure 16: GDP per capita and alternative CO_2 emissions per capita.

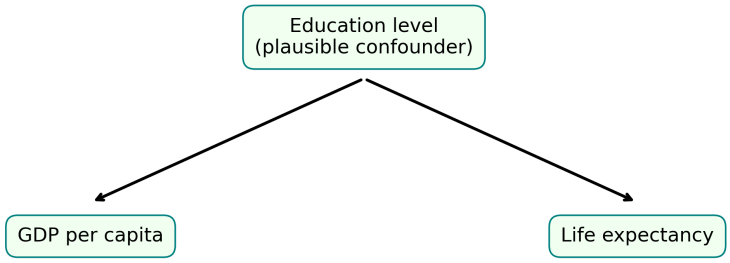
9 Confounding and Limits of Interpretation

For the GDP–life-expectancy relationship, let GDP per capita be X, life expectancy Y and education level Z. Education is a plausible common influence because it may be associated with productivity and earnings, while also relating to health knowledge and use of health services. The proposed diagram directs arrows from education to both outcomes.

$$X \leftarrow Z \rightarrow Y.$$

If education influences both variables, part of their observed association could reflect this common influence rather than a direct effect of GDP. Omitting it may therefore distort a causal interpretation. The diagram proposes a mechanism; it neither verifies that mechanism nor estimates its magnitude. Secondary gross enrollment is not identical to education level and would be only an imperfect proxy.

No adjustment for education has been performed. Other omitted factors, reverse relationships and common trends are also possible. The analysis establishes co-movement within the observed period but cannot identify the effect of changing GDP while holding other influences fixed. Advanced causal estimators are outside this investigation's scope.



Conceptual diagram: proposed relationships, not estimated causal effects.
This diagram shows a possible common cause; it is not a complete causal model.

Figure 17: Education as a plausible common influence on GDP per capita and life expectancy. Arrows are conceptual, not estimated effects.

10 Conclusion

Nepal's saved observations show broad increases in output per person, service coverage and Internet use, alongside substantial fluctuations in annual GDP growth. Dispersion comparisons depend on scale: GDP per capita has the largest raw numerical spread among four selected indicators, while Internet use has the largest relative spread. Forest area has no recorded variation in its available observations.

Strong positive correlations connect GDP per capita with the three required comparison indicators, but temporal trends and plausible confounding prevent causal conclusions. The unusual 2020 contraction is retained and investigated, and missing values remain explicit. These choices preserve the evidence rather than modifying it to produce cleaner patterns.

Reproducible calculation and defensible generalisation are distinct achievements. The descriptive results apply directly to the recorded years; the illustrative intervals and tests require additional assumptions and course clarification. Final submission must also resolve acceptance of the emissions substitution. Together, these qualifications define what the investigation establishes and where further evidence or methodological guidance is necessary.

References

[1] World Bank, "World development indicators," https://datacatalog.worldbank.org/search/dataset/0037712/world-development-indicators, 2026, accessed: 2026.

[2] ——, "Export competitiveness key to Nepal's green, resilient, and inclusive recovery, states new World Bank report," Press release, 12 April. https://www.worldbank.org/en/news/pressrelease/2021/04/12/export-competitivenesskey-to-nepal-s-green-resilient-and-inclusive-recovery-states-new-world-bank-report, 2021.

A Variable Dictionary and Coverage

Table 6 identifies the series used. The definitions below are explanatory summaries. The project-generated variable dictionary should accompany the submission with the exact downloaded definitions, source details and download timestamps. Those timestamps are not reconstructed from the report-writing date.

Table 6: Indicator codes and coverage of the saved working dataset.

Indicator	Code used	N	Missing year
GDP per capita	`NY.GDP.PCAP.KD`	15	None
GDP growth	`NY.GDP.MKTP.KD.ZG`	15	None
Life expectancy	`SP.DYN.LE00.IN`	15	None
Basic drinking water	`SH.H2O.BASW.ZS`	15	None
Secondary enrollment	`SE.SEC.ENRR`	14	2018
Electricity access	`EG.ELC.ACCS.ZS`	15	None
Internet use	`IT.NET.USER.ZS`	15	None
Alternative CO_2	`EN.GHG.CO2.PC.CE.AR5`	15	None
Forest area	`AG.LND.FRST.ZS`	14	2024

Assigned CO_2 code: `EN.ATM.CO2E.PC`. The alternative is labelled throughout and requires instructor acceptance. Potential observations: 15 per indicator.

Table 7: Measurement summaries and units.

Indicator	Measurement and interpretation
GDP per capita	Economic output per person, expressed in constant 2015 US dollars. It is not a measure of household income distribution.
GDP growth	Annual percentage change in real GDP. Negative observations indicate contraction.
Life expectancy	Expected years of life at birth under the mortality conditions represented by the indicator.
Basic drinking water	Percentage of the population using at least basic drinking-water services.
Secondary enrollment	Gross secondary-school enrollment as a percentage of the official secondary-school-age population.
Electricity access	Percentage of the population with access to electricity; reliability is not directly measured.
Internet use	Percentage of individuals using the Internet; connection quality is not directly measured.
Alternative CO_2	CO_2 emissions excluding LULUCF per capita, recorded in t CO_2e per capita.
Forest area	Forest area as a percentage of land area; forest condition is not directly measured.

These are explanatory summaries rather than verbatim metadata definitions. Consult the accompanying variable dictionary for exact source definitions.

B Investigation of the 2020 GDP-Growth Observation

The saved series records GDP growth of -2.3696% in 2020. It is the minimum across the 15 years and falls below the lower boxplot fence:

$$Q_1 - 1.5\,\mathrm{IQR} = 3.4735 - 1.5(2.3479) \approx -0.0484.$$

The upper fence is approximately 9.3433%. The observation is consequently flagged, but flagging is a prompt for investigation rather than a deletion rule.

Table 8: GDP growth surrounding the unusual 2020 observation.

Year	GDP growth (%)	Outside full-period IQR fences?
2018	7.6224	No
2019	6.6571	No
2020	-2.3696	Yes
2021	4.8382	No
2022	5.6313	No

Source: saved WDI working dataset. Fences use all 15 available growth observations, not only this five-year window.

The World Bank's release of 12 April 2021 describes Nepal's contraction in FY2020 and subsequent recovery in the context of the COVID-19 pandemic (2). This provides substantive support for interpreting the observation as consistent with a genuine contraction.

The release uses fiscal-year language and an earlier publication vintage. It supplies contextual evidence, not independent verification of the exact value in the saved WDI extract. Nor does it identify the causal contribution of individual factors from these data.

Decision: retain the original observation without alteration. The mean, variance, confidence interval and growth test all retain it.

C Data Quality and Reproducibility

The working dataset contains 15 unique years from 2010 to 2024, with Nepal and NPL as the country identifiers. The two missing observations are reported explicitly. Population-share indicators are checked against 0–100 bounds, while gross enrollment is not incorrectly constrained to that upper limit.

The calculation files have the following roles:

1. `01_data_download_and_quality.ipynb`: download records, raw-response preservation, missingness checks, metadata and the working dataset.

2. `02_descriptive_and_graphical_analysis.ipynb`: descriptive statistics, all required graphs, heterogeneity, covariance, correlation and the unusual-observation investigation.

3. `03_confidence_intervals_and_hypothesis_tests.ipynb`: illustrative intervals, both tests, assumptions and the combined test summary.

Analysis notebooks read the saved working CSV. Reproducing these results does not require replacing it with a fresh live download, which could contain revised values. Tables retain full numerical precision in CSV form, while figures are saved under `results/figures/`.

Table 9: Calculation outputs corresponding to report sections.

Report component	File under `results/tables/`
Descriptive statistics	`descriptive_statistics.csv`
Selected extremes	`selected_indicator_extremes.csv`
Dispersion	`heterogeneity_comparison.csv`
Dispersion rankings	`heterogeneity_measure_ranks.csv`
Boxplot calculations	`boxplot_outlier_summary.csv`
Covariance and correlation	`correlation_summary.csv`
Unusual observation	`unusual_observation_GDP_growth_2020.csv`
Confidence intervals	`illustrative_t_confidence_intervals.csv`
Hypothesis tests	`illustrative_hypothesis_tests_summary.csv`
Annual paired differences	`paired_access_differences.csv`